NEWS TOPIC CLASSIFICATION USING CLASSICAL MACHINE LEARNING: SPARSE FEATURES VS CONTEXTUAL EMBEDDINGS

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Statistical Machine Learning Project

ABSTRACT

This project explores multi-class news topic classification using classical machine learning techniques on different textual feature spaces. Experiments were conducted on the AG News dataset using Bag-of-Words, TF-IDF, and pretrained BERT embeddings. Dimensionality reduction techniques such as Truncated SVD and PCA were applied to analyze their effect on classification performance and visualization quality. Multiple machine learning models including Logistic Regression, SVM, KNN, and KMeans were evaluated using Accuracy, Weighted F1 and Macro F1-score.

***K*eywords** : NLP, News Classification, TF-IDF, BERT, PCA, SVD, t-SNE

1 Introduction and Dataset Used

News topic classification is a fundamental Natural Language Processing (NLP) task where textual news articles are categorized into predefined classes. Traditional NLP systems relied heavily on sparse statistical representations such as Bag-of-Words and TF-IDF. Recently, contextual embeddings generated using transformer models such as BERT have significantly improved text representation quality.

The objective of this project is to compare classical machine learning models across sparse and dense feature spaces while analyzing the impact of dimensionality reduction techniques such as Truncated SVD and PCA.

The AG News dataset from HuggingFace was used for all experiments. The dataset contains four news categories:

- World
- Sports
- Business
- Sci/Tech

For computational efficiency, a subset of the dataset was selected:

- Training samples: 8000
- Testing samples: 2000

2 Dataset Description and EDA

Text preprocessing was performed using built-in functionality from CountVectorizer and TfidfVectorizer.

The preprocessing pipeline included:

- Tokenization
- Stop-word removal
- Vocabulary pruning using `max_features`

The following exploratory visualizations were generated:

- Class distribution plot
- Text length distribution
- Sample text inspection

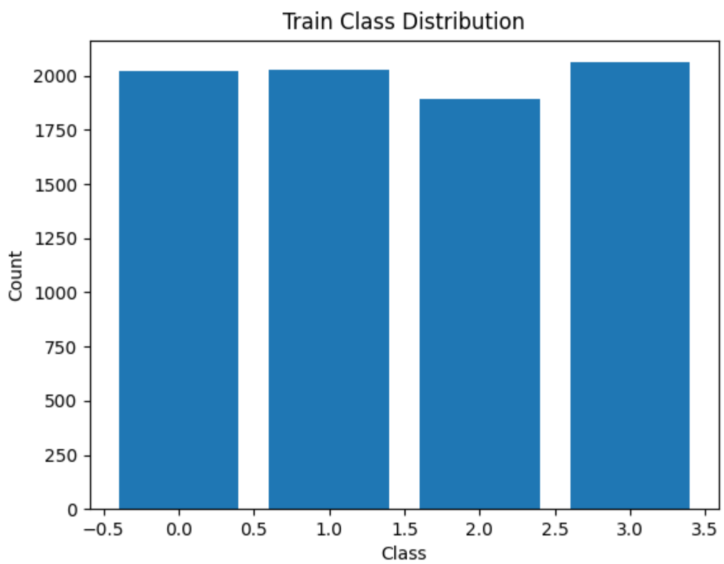


Figure 1: Class Distribution of AG News Dataset

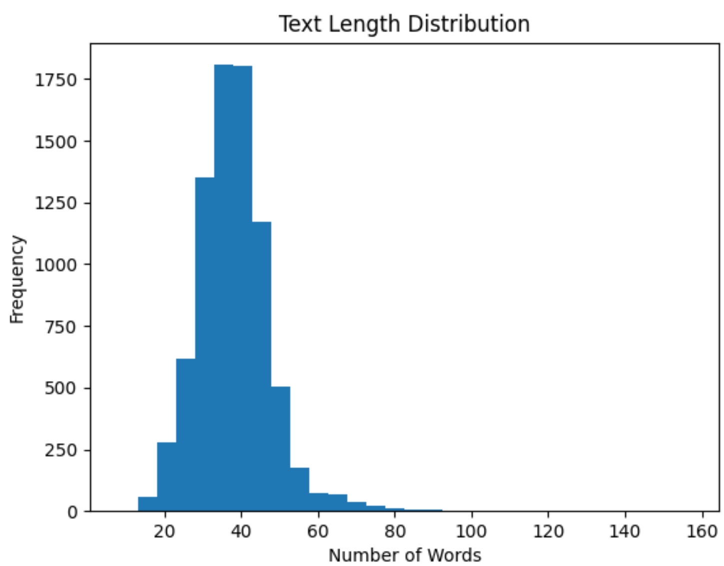


Figure 2: Distribution of Text Lengths

3 Methodology

3.1 Bag-of-Words

Bag-of-Words (BoW) was implemented using CountVectorizer with a vocabulary size of 5000.

3.2 TF-IDF

TF-IDF features were generated using TfidfVectorizer with unigram and bigram features:

- ngram range: (1,2)
- maximum features: 8000

3.3 BERT Embeddings

Pretrained BERT CLS token embeddings were extracted using a frozen transformer encoder. These embeddings provide dense contextual representations of the news articles.

3.4 Dimensionality Reduction

The following dimensionality reduction methods were applied:

- Truncated SVD on TF-IDF features
- PCA on BERT embeddings

Both methods reduced the feature dimensionality to 100 components.

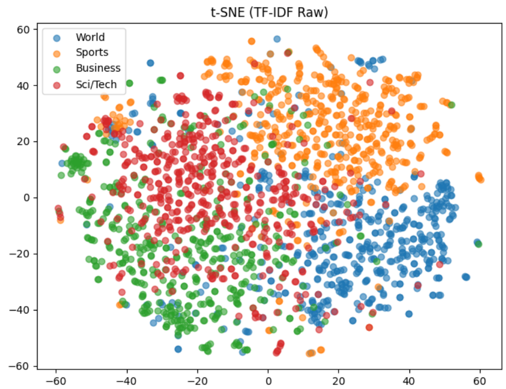
3.5 Models Used

The following classical machine learning models were evaluated:

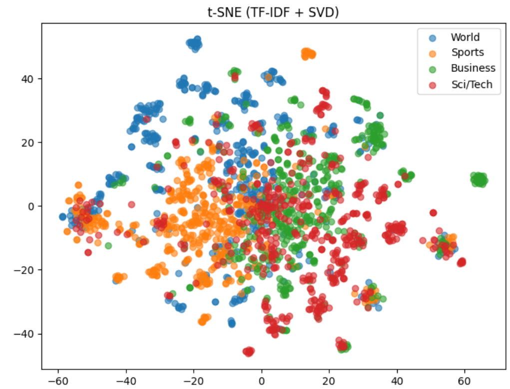
- Logistic Regression
- Linear SVM
- RBF SVM
- KNN
- KMeans

4 Experiments

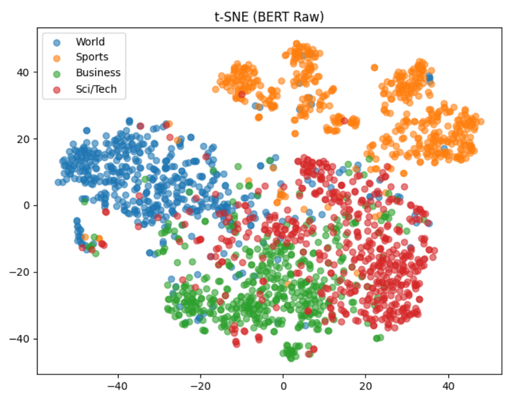
The t-SNE algorithm was used to visualize high-dimensional feature spaces in two dimensions.



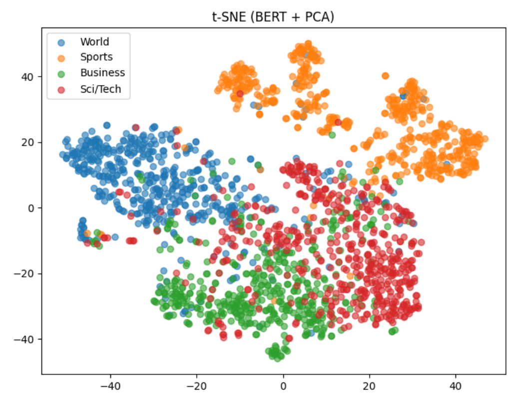
(a) TF-IDF



(b) TF-IDF + SVD



(c) BERT Raw

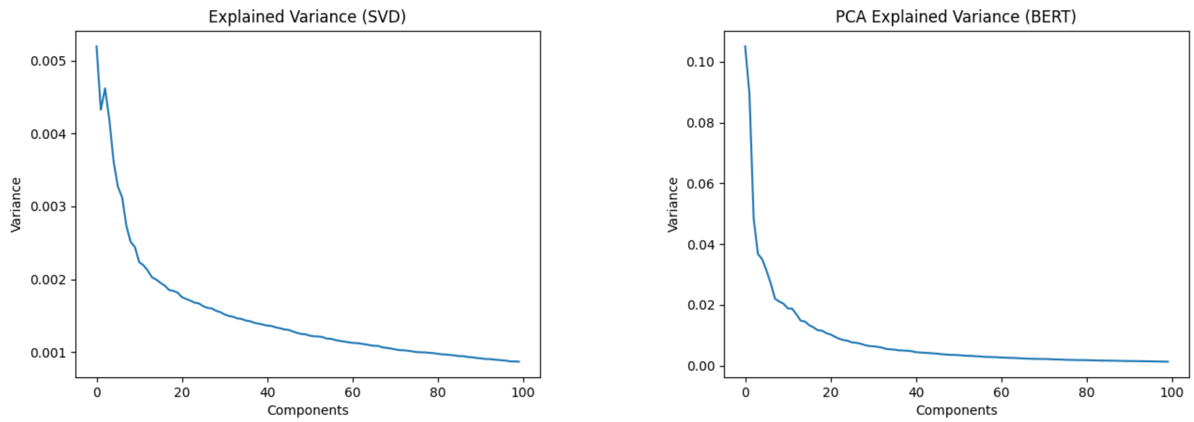


(d) BERT + PCA

Figure 3: t-SNE Visualizations of Different Feature Spaces

The BERT embeddings demonstrated better cluster separation compared to sparse TF-IDF features, indicating improved semantic understanding.

4.1 Explained Variance Analysis



(a) SVD Variance

(b) PCA Variance

Figure 4: Explained Variance Plots

The variance plots indicate that dimensionality reduction retained a significant amount of information while reducing computational complexity.

5 Results

5.1 Accuracy Comparison

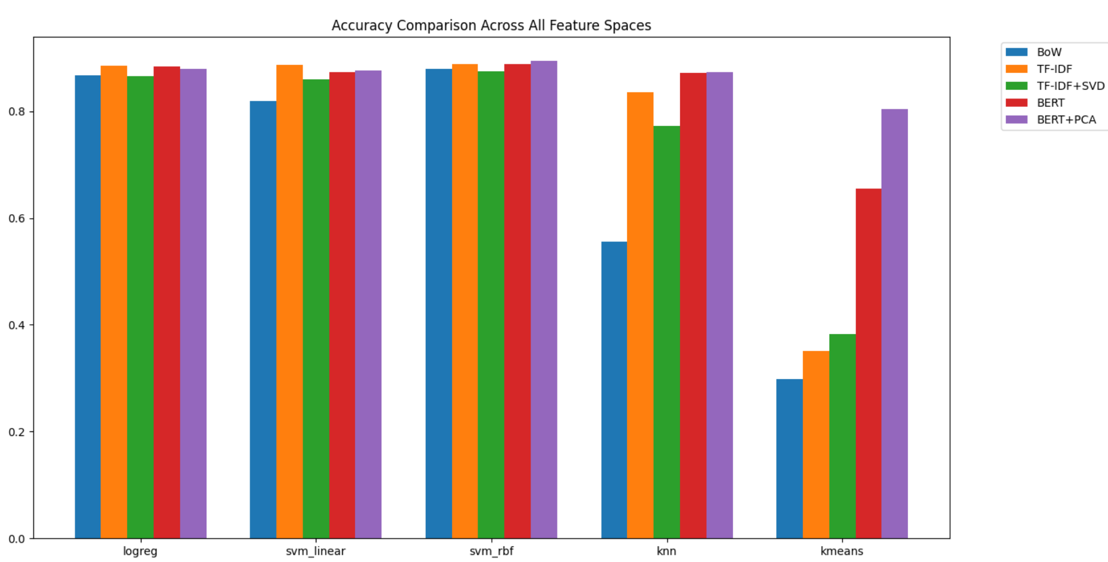


Figure 5: Accuracy Comparison Across Feature Spaces

5.2 Macro F1 Comparison

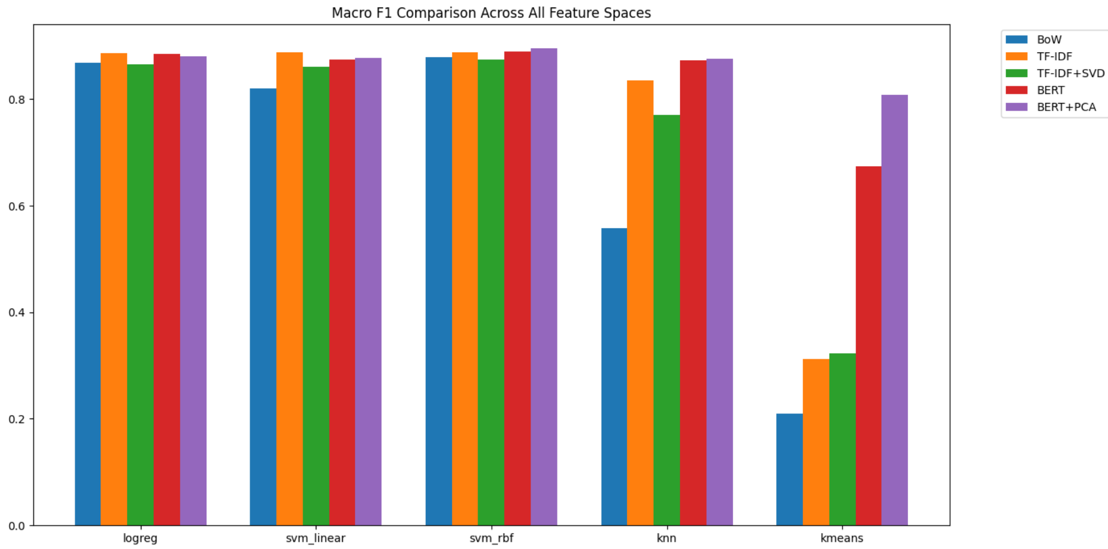


Figure 6: Macro F1 Comparison Across Feature Spaces

5.3 Confusion Matrix

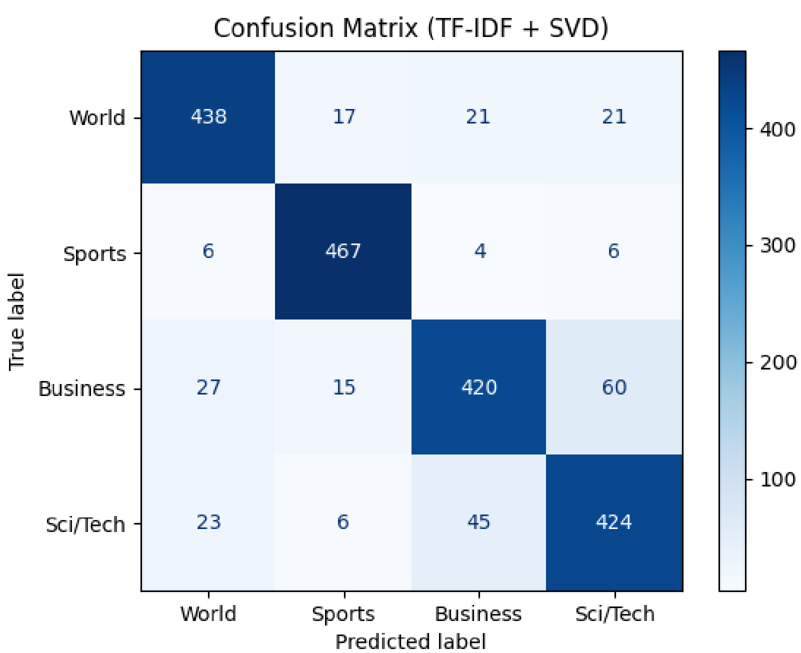


Figure 7: Confusion Matrix for TF-IDF + SVD using RBF SVM

6 Discussion

The experiments demonstrate that contextual embeddings generated using BERT outperform traditional sparse representations in most classification tasks.

Key observations from the experiments include:

- BERT embeddings produced more compact and semantically meaningful clusters.
- SVD reduced TF-IDF dimensionality while maintaining competitive classification performance.
- PCA reduced embedding dimensionality with minimal performance degradation.
- RBF SVM consistently achieved strong performance across feature spaces.
- KMeans performed worse due to its unsupervised nature.

7 Conclusion and Learnings

This project compared sparse textual representations and contextual embeddings for news topic classification using classical machine learning models.

The study showed that:

- Contextual embeddings provide richer semantic information than sparse representations.
- Dimensionality reduction techniques effectively reduce computational complexity while preserving important information.
- Classical machine learning models can still achieve strong performance when paired with high-quality feature representations.

Learnings from this project include:

- Understanding sparse and dense textual feature representations.
- Applying dimensionality reduction methods such as SVD and PCA.
- Visualizing high-dimensional data using t-SNE.
- Comparing supervised and unsupervised machine learning models.
- Evaluating NLP systems using Accuracy and Macro F1-score.

Acknowledgments

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